

Hospital Readmission Fairness Audit

DATASET: DIABETES 130-US PROTECTED ATTRS: RACE · GENDER · AGE

MODEL: LOGISTIC REGRESSION + RF NOT IN ANY EXAMPLE PROJECT

DATASET

Diabetes 130-US Hospitals (1999–2008)

UCI ML Repository · ~101,000 records · No credentialing required · Publicly available

race (protected)

gender (protected)

age (protected)

readmitted <30 days (target)

num_medications

num_procedures

time_in_hospital

num_diagnoses

discharge_disposition

Binary target: readmission within 30 days (1) vs not (0). Drop "readmitted = >30" rows or collapse into 0 – justify your choice in the data section.

STRUCTURE – BASED ON EXAMPLE PROJECT PATTERNS

SECTION

1

Introduction & problem framing

- Why hospital readmission prediction matters (CMS penalty programs, value-based care)
- Who gets harmed by a biased readmission model – specifically: a false negative means a high-risk patient is sent home without follow-up and may die or deteriorate
- State your research questions: does the model perform equally across racial groups? Across age groups? What happens when we debias?

Grader tip: every strong example opens with a concrete harm scenario, not just "fairness is important"

SECTION

2

Data exploration & bias analysis

- Class distribution of readmission by race, gender, age group
- Representation bias: how many records per group? Flag underrepresented groups
- Correlation analysis: do any features act as proxies for race? (e.g. discharge_disposition, payer_code)
- Historical bias: labels come from institutional discharge decisions – those may already encode prior discrimination

All 5 example projects have this section. Skipping it is a red flag to graders

SECTION

3

Baseline model & fairness metrics

- Train logistic regression baseline (matches what most example projects use – easy to compare)
- Report accuracy, AUC per group – not just overall
- Compute: demographic parity, equalized odds (FPR + FNR per group), equal opportunity (FNR only)
- Argue why FNR is the most important metric here – a missed high-risk patient is the costliest error

Justify your metric choice – Example 3 (mortgage) does this well, it's clearly expected

SECTION

4

Debiasing intervention

- **Pre-processing:** reweighting (Fairlearn's ExponentiatedGradient) or removing proxy features
- **In-processing:** fairness-constrained logistic regression via Fairlearn
- **Post-processing:** threshold adjustment per demographic group
- Pick at least two – compare them. Show what each one fixes and what it breaks

Examples 1, 2, 3 and 5 all apply at least one intervention – it's non-negotiable

SECTION

5

Fairness-accuracy trade-off

- Plot the Pareto frontier: fairness constraint strength on x-axis, accuracy drop on y-axis
- This is the centrepiece of every strong example project – don't skip it
- Discuss: is there a point where you gain fairness cheaply? Or does accuracy

collapse immediately?

SECTION

6

Ethics & accountability discussion

- Who deployed this model, and who is accountable when it fails?
- Asymmetry of costs: false positive (unnecessary readmission flag) vs false negative (patient sent home and deteriorates)
- Structural bias: payer_code (insurance type) as a proxy – links hospital AI to insurance discrimination
- Deployment context: should this model exist at all, or does it entrench existing disparities in care quality?

This is where your project separates from a technical exercise. Be specific – avoid vague "AI has risks" language

FAIRNESS METRICS TO REPORT

METRIC	WHAT IT MEASURES	WHY IT MATTERS HERE
Demographic parity	Equal positive prediction rates across groups	Baseline check – are some groups predicted high-risk at very different rates?
Equal opportunity (FNR)	Equal false negative rates across groups	Most critical here – unequal FNR means one group's sick patients are sent home more often
Equalized odds	Equal FPR and FNR simultaneously	Stronger criterion – show whether it's achievable and at what accuracy cost
Calibration	Predicted probability = actual reoffend rate	Optional but adds depth – does a score of 0.7 mean the same thing for all racial groups?

WHAT MAKES THIS PROJECT STAND OUT

- ◆ **Not in any of the 5 example projects.** Zero overlap with Adult dataset (Ex.1), Communities & Crime (Ex.2), mortgage loans (Ex.3), CV detection (Ex.4), or student dropout (Ex.5). Graders will notice immediately.
- ◆ **Three intersecting protected attributes.** Race, gender, and age all interact. You can do an intersectional analysis (e.g. elderly Black women vs young white men) – exactly the approach Example 1 used to great effect, but in a higher-stakes setting.
- ◆ **Insurance as a proxy feature.** payer_code encodes insurance type, which correlates strongly with race and socioeconomic status. Analysing whether removing it improves fairness – and what it costs in accuracy – is a concrete, original contribution.
- ◆ **The ethics section writes itself.** Hospital readmission models are deployed right now in US hospitals under CMS programs. You can point to real deployed systems, real penalties, and real documented disparities in readmission rates by race.

ETHICS SECTION – WHAT TO ARGUE

The core argument: a false negative in this context is not a neutral technical error. It means a clinician or discharge coordinator trusts the model's low-risk prediction, sends the patient home, and that patient deteriorates without follow-up care. If your model produces higher FNRs for Black patients or elderly patients, it is not just statistically unfair – it is producing worse medical outcomes for already-disadvantaged groups.

Accountability gap: Who is liable when the model is wrong? The hospital? The software vendor? The clinician who relied on it? The EU AI Act and the US CMS program create partial accountability structures, but neither directly assigns liability for biased predictions. Discuss this gap concretely.

The structural link: payer_code (insurance type) connects hospital AI to insurance discrimination. A model trained on data where uninsured patients received less follow-up care will learn to predict low-risk for uninsured patients – because they were discharged and not readmitted, not because they were healthier. This is a classic label bias problem and worth naming explicitly.

KEY PAPERS TO CITE

Strack et al. (2014) – Impact of HbA1c Measurement on Hospital Readmission Rates

Original paper introducing the Diabetes 130-US dataset. Cite for dataset description.

Obermeyer et al. (2019) – Dissecting racial bias in an algorithm used to manage the health of populations

Science. The landmark paper on racial bias in a commercial healthcare algorithm. Essential background for your ethics section.

Hardt et al. (2016) – Equality of Opportunity in Supervised Learning

Defines equalized odds and equal opportunity. Cite when introducing your fairness metrics. Used in Example 1.

Mehrabi et al. (2021) – A Survey on Bias and Fairness in Machine Learning

Good taxonomy of bias types (representation bias, label bias, etc.) for your data section.

Chouldechova (2017) – Fair prediction with disparate impact

Impossibility theorem. Optional but adds theoretical depth if you want to discuss the fairness-accuracy trade-off formally.

RECOMMENDED TOOLS

PYTHON LIBRARIES

- fairlearn – debiasing + metrics
- sklearn – models + preprocessing
- pandas / numpy – data wrangling
- matplotlib / seaborn – plots
- shap – optional: feature importance

KEY FAIRLEARN CLASSES

- MetricFrame – per-group metrics
- ExponentiatedGradient – in-proc.
- ThresholdOptimizer – post-proc.
- demographic_parity_difference
- equalized_odds_difference

One thing to decide early: how to handle the "readmitted >30 days" group. You can (a) drop them and only classify <30 vs not-readmitted, or (b) collapse >30 into "not readmitted" and do a binary task. Option (a) is cleaner and more defensible – document the choice explicitly. The graders in Example 3 and Example 5 both flagged label construction as a key methodological decision.

